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Youth coping and symptoms of anxiety and depression: associations with age, gender, and peer stress

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Abstract

During late childhood and early adolescence, youth experience heightened stress exposure and rates of mental health disorders, emphasizing the need to investigate how youth cope with stress as a potential mechanism of risk and/or resilience. This study examined youths' age, gender, and peer-related stress exposure as moderators of the relationships among coping and anxious/depressed symptoms during this important developmental period. To sample participants with a broad range of internalizing psychopathology symptoms, a community sample of 120 older children and adolescents (46% female, M age = 12.28) was recruited through mental health clinics and a university-based study finder in a southeastern metropolitan area. Participants completed a battery of questionnaires assessing coping strategies, stress exposure, demographic information, and anxious/depressed symptoms in a cross-sectional study. Analyses included interactions among age, gender, and peer stress as predictors of youths' coping strategies and anxious/depressed symptoms. There were significant associations between age, gender, coping strategies, and symptoms, where girls reported more anxiety and depression symptoms compared to boys, and primary and secondary control coping strategy use increased with age. The findings expand upon previous studies by showing gender differences in the relation among secondary control coping and internalizing problems at different ages. Additionally, findings demonstrate that

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Ethics Approval. Approval was obtained from the Vanderbilt University Institutional Review Board (IRB # 181531). The procedures used in this study adhere to the tenets of the Declaration of Helsinki.

Consent to participate. Informed consent was obtained from all individual participants included in the study.

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degrees of peer stress exposure show associations with disengagement coping across development, which may contribute to the onset of internalizing symptoms in older girls. Findings may be used to inform interventions by promoting the use of more adaptive forms of coping to prevent internalizing disorders in youth. Limitations and directions for future research are discussed.

Keywords

age; gender; coping; stress; internalizing

INTRODUCTION

In emerging adolescence, peer-related stress increases along with the risk of developing internalizing problems. There are also well-established gender differences in rates of reported peer stress, anxiety, and depression during adolescence, with generally higher rates for girls (Hankin et al., 2009; Rutter et al., 2003; Zahn-Waxler et al., 2006). Importantly, the ways in which youth cope with stress may moderate the association between peer stress and psychopathology (Compas et al., 2017; Eisenberg et al., 2010), and adolescents experience important changes in cognitive and emotional development that support the understanding of increasingly complex coping strategies (Compas et al., 2017). In contrast, it is unclear whether these changes in cognitive ability are related to adolescents' *use* of more effective coping strategies. Furthermore, while multiple theories have helped to explain gender differences in internalizing problems, existing studies have varied significantly in their methods and ages sampled, making findings difficult to integrate. Given heightened rates of internalizing psychopathology across adolescence, particularly among girls, clarifying these associations has significant implications for intervention and prevention efforts. To address several gaps in this area of research, the present study used a control-based model of coping (Compas et al., 2017; Connor-Smith et al., 2000; Rudolph et al., 1995; Weisz et al., 1994) to examine gender and age differences in coping with peer stress and associations with symptoms of anxiety and depression in youth.

Peer Stress and Risk for Internalizing Disorders

During emerging adolescence, peer relationships are a central component of interpersonal functioning as youth become increasingly engaged in interactions with others outside the family (Bukowski et al., 2011; Persike & Seiffge-Krenke, 2016; Sontag et al., 2008; Wong et al., 2020). Stress that arises in interpersonal relationships is particularly salient, as 80% or more of daily stressors reported by youth are within the interpersonal domain (Calvete et al., 2011; Sieffge-Krenke & Persike, 2017; Seiffge-Krenke et al., 2009). This ongoing stress in peer relationships is related to increased risk for anxiety and depressive disorders in youth (Grant et al., 2003; Lee & Hankin, 2009; Young et al., 2019; Zandstra et al., 2015; Zimmer-Gembeck, 2017). Further, there are well-established gender and age differences in internalizing disorder prevalence across development, with generally higher rates for girls (e.g., Hankin et al., 2009; Rutter et al., 2003; Zahn-Waxler et al., 2006). Gender differences in depression appear between ages 12–14, diverging significantly in middle adolescence (Dalsgaard et al., 2020; Salk et al., 2017; Twenge & Nolen-Hoeksema, 2002). Gender differences in anxiety disorders follow a similar course, with higher rates of

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anxiety disorders emerging in mid-adolescence (Dalsgaard et al., 2020). Existing research emphasizes the roles of peer stress and the coping strategies that adolescents use to help explain these age and gender differences in internalizing problems (e.g., Cyranowski et al., 2000; Hankin et al., 2007). However, findings have varied significantly and research outlining how age and gender correspond to patterns of coping is less clear. Better understanding the mechanisms underlying age and gender differences in coping and psychopathology may help inform more targeted interventions for at-risk youth.

Coping and Risk for Psychopathology: Age Considerations

A strong body of evidence supports coping as a significant moderator between stress and youth psychopathology (Compas et al., 2017; Eisenberg et al., 2010). Coping can be defined as "conscious volitional efforts to regulate emotion, cognition, behavior, physiology, and the environment in response to stressful events or circumstances" (Compas et al., 2001). The present study uses a well-validated control-based model of coping (e.g., Benson et al., 2011, Compas et al., 2006, 2017; Connor-Smith et al., 2000; Rudolph et al., 1995; Wadsworth & Compas, 2002; Weisz et al., 1994) that includes three coping factors: primary control coping, secondary control coping, and disengagement coping. Broadly referred to as engagement coping strategies, primary control coping includes actively changing a stressor or one's emotional reaction to the stressor (i.e., emotional expression, emotional modulation, and problem-solving), while secondary control coping entails more cognitively complex strategies (i.e., positive thinking, distraction, acceptance, and cognitive reappraisal) to adapt to a stressor. Disengagement coping strategies involve attempting to orient away from a stressor or one's emotions (i.e., wishful thinking, denial, avoidance) (Compas et al., 2017). These coping factors have shown robust associations with internalizing and externalizing symptoms in youth. For instance, primary control and secondary control coping strategies show negative associations with internalizing and externalizing problems (Compas et al., 2017), whereas disengagement coping shows a positive association with symptoms (Compas et al., 2001).

There is evidence for age differences in specific coping strategies during childhood and adolescence (e.g., Nolen-Hoeksema & Aldao, 2011; Skinner & Zimmer-Gembeck, 2007; Zimmerman & Iwanski, 2014). Across development, problem-solving (i.e., primary control coping) shows increased use, the use of distraction strategies (i.e., secondary control coping) is either stable or increases, and the use of avoidance (i.e., disengagement coping) decreases with age (Eschenbeck et al., 2018; Zimmer-Gembeck & Skinner, 2011). Importantly, because adolescence reflects a period of increased self-reliance and cognitive ability, coping strategies may become more active, diverse, and cognitively complex (Compas et al., 2017). Thus, it is likely that youth develop an increased capacity for using primary and secondary control coping strategies as they age, whereas disengagement coping use may decrease over time (Zimmer-Gembeck & Skinner, 2011). However, despite youths' ability to use more potentially effective forms of coping, they also encounter greater levels of peer stress, a salient risk factor for the onset of internalizing disorders (Al-Bahrani et al., 2013; Hankin et al., 2015; Skinner & Zimmer-Gembeck, 2007; Zimmer-Gembeck & Skinner, 2011). These high levels of peer stress may tax youths' coping resources (Anderson et al., 2021; Abaied & Rudolph, 2011; Watson et al., 2021), resulting in greater reliance on disengagement coping

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strategies and decreased use of more active and complex coping strategies (i.e., primary and secondary control coping). Thus, the types of coping strategies youth employ may be moderated by the levels of peer stress they encounter.

There may also be age differences in the relationship between youths' coping and symptoms of depression and anxiety. Within the control-based model, decreased primary and secondary coping strategy use has been shown to correspond to increased internalizing symptoms in youth (Compas et al., 2017). Further, age has emerged as a moderator of the relations among subtypes of coping (e.g., cognitive reappraisal, emotional suppression) and internalizing symptoms, such that relationships were stronger for adolescents than for children (Compas et al., 2017). However, findings on age-related differences in coping strategies and their association with symptoms have been challenging to integrate due to methodological and conceptual differences across studies (Eschenbeck et al., 2018; Zimmer-Gembeck & Skinner, 2011, 2017). For instance, seminal work on age differences in coping has focused on developmental (i.e., inherent, *stage-related* changes to an individual's coping as they age; Vaillant, 1977) or contextual models (i.e., changes in the types of *stressors* that individuals cope with as they age; Folkman et al., 1987; Folkman & Lazarus, 1980; McCrae, 1982), with findings varying significantly and fewer studies focusing on changes in coping during emerging adolescence. Further, relatively little research on age-related differences in coping has been based on the control-based model of coping. Understanding age-related differences in coping can provide a framework for expected and appropriate coping abilities, as well as highlight targets for intervention.

Coping and Risk for Psychopathology: Gender Considerations

Several theories have also been posited to explain gender differences in internalizing psychopathology, with some research emphasizing the roles of peer-related stress and gender differences in coping. Prior research suggests that adolescent girls experience greater exposure to interpersonal stressors, which has been posited to contribute in part to higher rates of internalizing problems during adolescence (Hankin et al., 2007). While findings related to gender differences in coping show inconsistent findings, several studies demonstrate that girls use more primary control coping strategies (e.g., support seeking, problem-solving) relative to boys (e.g., Calvete et al., 2011; Eschenbeck et al., 2018; Nolen-Hoeksema & Aldao, 2011; Persinke & Sieffge-Krenke, 2012; Zimmerman & Iwanski, 2014). Some studies have shown that across age groups, boys use more disengagement coping strategies (e.g., Connor-Smith et al., 2000; Eschenbeck et al., 2007; Eschenbeck et al., 2018; Zimmermann & Iwanski, 2014), while others have found greater use of these strategies in girls (e.g., Al-Bahrani et al., 2013; Calvete et al., 2011; Hampel & Petermann, 2005). Nolen-Hoeksema and Aldao (2011) found that women tend to use more secondary control coping, while other studies have demonstrated that boys use more secondary control coping strategies (e.g., Connor-Smith et al., 2000; Hampel & Petermann, 2005). Inconsistencies in findings may be attributable to disparate age groups and measures across studies.

Few studies have integrated this research to evaluate the interaction of age and gender in association with youth coping and internalizing problems. Extant research indicates that

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gender differences in coping (i.e., increased primary control coping use for girls) may be more evident during adolescence relative to childhood (Eschenbeck et al., 2007). This potential gender difference in coping in adolescence coincides with a heightened risk for internalizing disorders that occurs with age, particularly for girls (Dalsgaard et al., 2020; Hankin & Abramson, 2001). Still, gender differences in the effectiveness of distinct types of coping throughout adolescence are not well understood.

Summary and Goals for Present Study

In summary, the present study focuses on three areas of research that require further investigation to advance understanding of coping with stress in late childhood and early adolescence: (1) examining age and gender differences in coping and internalizing symptoms, (2) investigating peer stress as a potential moderator of these associations, and (3) assessing how age and gender interact with youths' coping in association with their symptoms.

Hypotheses.—First, given findings that primary and secondary control coping use tends to increase as youth age (e.g., Compas et al., 2017; Zimmer-Gembeck & Skinner, 2011), while disengagement coping use decreases with age (Skinner & Zimmer-Gembeck, 2007; Zimmer-Gembeck & Skinner, 2011), it was hypothesized that age would show a positive association with the use of primary (Hypothesis 1a) and secondary control coping (Hypothesis 1b), but a negative association with disengagement coping use (Hypothesis 1c). Second, it was hypothesized that greater levels of peer stress would show a negative association with primary (Hypothesis 2a) and secondary control coping use (Hypothesis 2b), but a positive association with disengagement coping use (Hypothesis 2c). Third, it was expected that age would interact with peer stress, where older youth with higher levels of peer stress would use more disengagement coping (Hypothesis 3). Fourth, based upon findings that girls use more primary control coping strategies compared to boys (e.g., Calvete et al., 2011; Eschenbeck et al., 2018; Nolen-Hoeksema & Aldao, 2011; Persinke & Sieffge-Krenke, 2012; Zimmerman & Iwanski, 2014), and that coping-related gender differences may be more salient later in development (Eschenbeck et al., 2007), it was hypothesized that older girls would use more primary control coping (Hypothesis 4). Fifth, given stronger associations between coping and symptoms in adolescents relative to children (Compas et al., 2017), it was hypothesized that age and coping would interact, where older youth who use more disengagement coping would have higher symptoms (Hypothesis 5). Finally, due to mixed findings in prior research, gender was examined as a potential moderator between disengagement coping and symptoms. Interactions between gender and age with primary and secondary control coping were also exploratory.

METHOD

Participants

The full sample was comprised of 120 older children and adolescents drawn from a larger study on how mothers and their children communicate about emotions and stress. In efforts to enroll participants with variability in symptoms of psychopathology, participants from a community population were recruited through the Family Care Partners Database, a

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university-based study finder, as well as private and public mental health clinics. Data on one or more key variables was missing for three participants. These three participants did not differ significantly from the 117 included youth on the key study variables (i.e., peer stress, age, gender, coping, and anxious/depressed symptoms) and were, therefore, listwise deleted. Excluded participant data accounted for 1.46% of all study data. Little's test confirmed that data were missing completely at random (Little, 1988) ($ps > .05$).

The final sample for analyses included 117 older children and adolescents (46% female), ages 9 to 15 ($M = 12.28$, $SD = 1.89$). The sample included 10 9-year-olds (18.5%), four 10-year-olds (7.4%), seven 11-year-olds (13.0%), six 12-year-olds (11.1%), 11 13-year-olds (20.4%), 11 14-year-olds (20.4%), and five 15-year-olds (9.3%). Participants' race and ethnicity were 67.2% White, 25.2% Black or African American, 5.1% Latino or Hispanic, and 2.5% Asian. Annual income for participants' families ranged from < $10,000 to > $300,000, (median income = $68,000).

Measures

Youth Peer Stress.—The Responses to Stress Questionnaire-Peer Stress version (RSQ-PS; Connor-Smith, et al., 2000) was used to measure youths' exposure to stressors in peer relationships. Designed for youth ages 9 and up, the RSQ-PS is self-report measure including 10 items depicting aspects of social life, including peer pressure (e.g., "Being pressured by others to do something"), romantic relationships (e.g., "Asking someone out and being turned down"), social exclusion (e.g., "Being left out or rejected"), friendships (e.g., "Having problems with a friend"), and bullying (e.g., "Being teased or hassled by other kids"). Youth indicated how stressful relevant experiences were over the last six months on a 4-point Likert scale (1 = *not at all*, 2 = *a little*, 3 = *somewhat*, 4 = *very*). Peer stress items on the RSQ-PS have been used reliably in prior research (e.g., Henry et al., 2020; Watson et al., 2021), and internal consistency for the current sample was good, $\alpha = .86$.

Youth Coping.—The RSQ-PS also includes 57 items to examine individuals' coping strategies in response to peer-related stress, also rated on a 4-point Likert Scale (0 = *not at all*, 1 = *a little*, 2 = *some*, 3 = *a lot*). Analyses focused on the three coping factors validated in multiple factor analytic studies (e.g., Compas et al., 2006; Connor-Smith et al., 2000): primary control, secondary control, and disengagement coping. In efforts to account for any individual differences in base rates of item endorsement and potential response bias, proportion scales were computed by dividing each coping factor's total score by the overall total coping score (e.g., Gruhn et al., 2019; Vitaliano et al., 1987). The RSQ-PS has consistently demonstrated strong validity and reliability and has been used in several studies examining youth coping and psychological adjustment (Anderson et al., 2021; Henry et al., 2020; Wadsworth et al., 2004; Watson et al., 2021; Yao et al., 2010). Youths' peer stress-related coping showed acceptable internal consistency reliabilities ($\alpha = .87$ on primary control coping, $\alpha = .86$ on secondary control coping, $\alpha = .76$ on disengagement coping).

Youth anxious/depressed symptoms.—The 13-item anxious/depressed subscale of the Youth Self Report was used to measure adolescents' symptoms of anxiety and depression (YSR; Achenbach & Rescorla, 2001). Children and adolescents in the present

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sample rated statements on a 3-point Likert scale (0 = *not at all true*, 1 = *somewhat true*, 2 = *very true*). Example items include: "I am afraid of going to school;" "I feel worthless or inferior;" and "I am nervous or tense." To maximize variance, raw scores were utilized across all analyses. Validity and reliability are well-established for the YSR (Achenbach & Rescorla, 2001; Gomez et al., 2014; Ivanova et al., 2007; Liu et al., 1997), and the empirically derived subscale of anxious/depressed symptoms has been well-validated in youth research (e.g., Ebesutani et al., 2010; Gomez et al., 2014; Ivanova et al., 2019; Ivarsson et al., 2002; Vasin et al., 2016). Internal consistency reliability for the anxious/depressed subscale of the YSR was $\alpha = .83$ in the present study. Although the YSR is specified for 11-to-18-year-olds, there were 26 youth in the present sample that were nine-10 years old. For this subgroup, internal consistency reliability was adequate, $\alpha = .74$.

Study Design

The current study used a cross-sectional and self-report design to test associations among youths' age, gender, and peer-related stress exposure in association with coping and anxious/depressed symptoms during an important developmental period. Youth were recruited via email through a university-based study finder, the Family Care Partners Database, and public and private mental health clinics throughout a large southeastern metropolitan area inviting them to participate in a study on how mother-child dyads communicate about coping, emotions, and stress. After families expressed interest, trained doctoral students contacted families and completed telephone screens to determine eligibility. Maternal history of bipolar I, bipolar II, schizophrenia, or youth history of pervasive developmental disorder or intellectual disability were exclusion criteria for the present study. Before the laboratory visit, youth were self-administered a battery of questionnaires with instructions included. Trained doctoral students screened surveys for endorsement of critical response items (e.g., suicidal ideation on the YSR) and to ensure the participants' comprehension and completion of surveys. Surveys ensured the anonymity of participants. During the laboratory visit, families provided informed consent for participation and completed a range of tasks and interviews that were not tested in the current study. The same procedure was used across all participants in the present study, regardless of child age or gender. Families were debriefed at the end of the laboratory visit, offered an opportunity to ask any additional questions, and compensated for their time. The University Institutional Review Board approved all study procedures.

Data Analytic Plan

Statistical analyses were conducted with SPSS (version 25). Preliminary analyses were used to replicate well-established age and gender differences in symptoms of depression and anxiety, as well as the interactions among gender, age, and peer stress exposure in association with symptoms. Bivariate Pearson correlations were used to provide an overview of the associations between youths' age, gender, stress, coping, and symptoms. Multiple linear regression analyses were used to test hypotheses. Separate models were computed to predict youths' primary control, secondary control, and disengagement coping from gender, age, and peer-related stress, including interactions between (1) age and gender and (2) age and peer stress. Additional models were conducted to account for variance in youths'

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anxious/depressed symptoms as a function of age, gender, and coping, including interactions of (1) age and coping and (2) gender and coping.

Power Analyses

Power analyses were conducted using G*Power 3.1 to identify which effect sizes could be detected in correlational and regression analyses (Erdfelder et al., 1996). With $\alpha = .05$ and power set to .80, power analyses indicated the ability to detect small-to-medium correlations ($r = .25$ or larger) and small-to-medium effects ($f^2 = .10$ - .15) in multiple regression analyses.

RESULTS

Descriptive Statistics and Preliminary Analyses

Descriptive statistics are shown in Table 1. There was a significant difference in anxious/depressed symptoms between boys (*M* raw score = 3.83, *SD* = 3.60) and girls (*M* raw score = 5.91, *SD* = 4.17), $t(115) = -2.90$, $p = .004$, such that girls reported higher symptoms. Girls ($M = 16.20$, $SD = 5.93$) and boys ($M = 15.93$, $SD = 5.80$) reported similar levels of peer stress, $t(115) = -.25$, $p = .85$.

The first set of preliminary regression analyses included age and gender as independent variables and youths' anxious/depressed symptoms as the dependent variable. The overall model was significant, $F(3, 114) = 6.68$, $p < .001$; adjusted $R^2 = .13$. Further, gender moderated the association between age and anxious/depressed symptoms ($\beta = .27$, $p = .003$), where the relation among age and symptoms was significant for girls ($b = .79$, $t(113) = 3.07$, $p = .003$) but not for boys ($b = -.33$, $t(113) = -1.24$, $p = .22$). The second set of preliminary regression analyses included age and peer stress as independent variables, youths' anxious/depressed symptoms as the dependent variable, and gender as a covariate. The overall model was significant, $F(5, 112) = 18.02$, $p < .001$, adjusted $R^2 = .37$. Further, youth who reported more peer stress had higher symptoms ($\beta = .55$, $p < .001$). The interaction between age and peer stress was significant ($\beta = .27$, $p = .04$), where at high levels of peer stress ($b = .49$, $t(113) = 2.31$, $p = .02$), older youth had higher symptoms. At low ($b = -.12$, $t(113) = -.53$, $p = .60$) and moderate levels of peer stress ($b = .19$, $t(113) = 1.18$, $p = .24$), age was unrelated to youths' symptoms.

Bivariate Correlational Analyses

Hypotheses 1 & 2: Correlations Among Key Variables.—The first and second hypotheses (i.e., associations among youths' age, peer-related stress, and coping) were tested using bivariate correlational analyses (Table 2). In support of Hypothesis 1b, age showed a significant positive correlation with secondary control coping ($r = .21$, $p = .02$), where greater secondary control coping use corresponded to older age. Conversely, the correlations between age and youths' use of primary control ($r = .16$, $p = .09$; Hypothesis 1a) and disengagement coping were not significant ($r = .12$, $p = .20$; Hypothesis 1c). In support of Hypotheses 2a/b, peer stress was significantly negatively correlated with youths' primary ($r = -.20$, $p = .03$) and secondary control coping use ($r = -.43$, $p < .001$). Peer stress and disengagement coping use were unrelated (Hypothesis 2c).

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Primary control ($r = -.20$, $p = .03$) and secondary control coping ($r = -.45$, $p < .001$) were negatively correlated with anxious/depressed symptoms, where youth reporting more primary and secondary control coping had fewer anxious/depressed symptoms. In contrast, disengagement coping ($r = .23$, $p = .01$) and peer stress ($r = .55$, $p < .001$) showed positive correlations with symptoms, where greater disengagement coping and peer stress were associated with increased symptoms.

Multiple Linear Regression Analyses: Predictors of Coping

Hypothesis 3: Predicting Coping from Age and Peer Stress.—To test the third hypothesis, age, and peer stress were included as independent variables, youths' coping was the dependent variable, and gender was a covariate (Table 3, Model 1). Separate models were conducted for primary, secondary, and disengagement coping. The overall model predicting secondary control coping was significant, $F(5, 112) = 9.65$, $p < .001$, adjusted $R^2 = .23$. There were main effects of both age ($\beta = 0.22$, $p = .01$) and peer stress ($\beta = -0.45$, $p < .001$) in predicting youths' use of secondary control coping. The overall model predicting primary control coping was significant, $F(5, 112) = 2.74$, $p = .03$, adjusted $R^2 = .06$. Peer stress showed a significant negative association with youths' primary control coping use ($\beta = -0.21$, $p = .03$). In the model predicting disengagement coping (Hypothesis 3), the interaction between age and peer stress exposure was significant ($\beta = 0.25$, $p = .01$), where at lower levels of stress exposure ($b = -0.01$, $t(112) = -2.73$, $p = .01$), older youth used significantly less disengagement coping strategies than younger youth (Figure 1). At moderate ($b = -0.002$, $t(112) = -1.43$, $p = .16$) and high levels of peer stress ($b = -0.002$, $t(112) = .75$, $p = .46$), age was unrelated to youths' use of disengagement coping.

Hypothesis 4: Predicting Coping from Age and Gender.—To test the fourth hypothesis, age and gender were independent variables, and youths' coping served as the dependent variable (Table 3, Model 2). Separate models were conducted for primary, secondary, and disengagement coping. The overall model predicting secondary control coping use was significant, $F(4, 113) = 3.71$, $p = .01$, adjusted $R^2 = .07$. Age showed a significant positive association with secondary control coping ($\beta = 0.23$, $p = .01$). Gender was a significant moderator of the association between age and secondary control coping ($\beta = -0.20$, $p = .03$), where the relation among age and secondary control coping was significant for boys ($b = 0.01$, $t(113) = 3.13$, $p = .002$) but not girls ($b = 0.00$, $t(113) = .12$, $p = .91$). The main effects for age, gender, and their interactions were not significant predictors of youths' use of primary control (Hypothesis 4) or disengagement coping.

In the full models including age, gender, and levels of peer stress as predictors of youths' coping, age showed a significant positive association with primary control coping use ($\beta = 0.20$, $p = .03$), such that older youth used more primary control coping strategies (Table 3, Model 3). Age and peer stress remained significant predictors of secondary control coping. Further, the interaction between age and peer stress predicting disengagement coping remained significant. However, gender was no longer a significant moderator of the association between age and secondary control coping. An exploratory three-way interaction of Age x Gender x Peer Stress predicting coping was also examined, however, it was not a significant predictor of any of the three coping factors.

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Multiple Linear Regression Analyses: Predictors of Anxious/Depressed Symptoms

Hypothesis 5: Predicting Symptoms from Age and Coping.—To test the fifth hypothesis, age and coping were independent variables, anxious/depressed symptoms were the dependent variable, and gender was a covariate (Table 4, Models 1–3). The models for all three coping factors were significant ($ps < .01$). Age showed a significant positive association with anxious/depressed symptoms in the secondary control coping model ($\beta = 0.23$, $p < .001$). There were main effects of primary control ($\beta = -0.26$, $p = .01$), secondary control ($\beta = -0.48$, $p < .001$), and disengagement coping ($\beta = 0.20$, $p = .02$) in association with symptoms. In support of Hypothesis 5, age moderated the association between disengagement coping and anxious/depressed symptoms ($\beta = 0.23$, $p = .01$). Namely, higher levels of disengagement coping corresponded to more symptoms in older youth ($b = 0.77$, $t(112) = 3.08$, $p = .003$), whereas age was unrelated to anxious/depressed symptoms when youth used moderate ($b = 0.27$, $t(112) = 1.45$, $p = .15$) or low levels ($b = -0.24$, $t(112) = -.89$, $p = .37$) of disengagement coping strategies (Figure 2).

Exploratory Analyses: Predicting Symptoms from Gender, Age, and Coping.—For exploratory analyses, gender and coping were independent variables, symptoms were the dependent variable, and age was a covariate (Table 4, Models 4–6). All three models were significant ($p < .001$). Gender showed significant associations with primary control ($\beta = 0.31$, $p < .001$), secondary control ($\beta = 0.24$, $p < .003$), and disengagement coping ($\beta = 0.26$, $p < .003$) models. Gender moderated the association between disengagement coping and anxious/depressed symptoms ($\beta = 0.60$, $p = .02$), where the relation among disengagement coping and symptoms was significant for girls ($b = 55.35$, $t(112) = 3.48$, $p = .001$) but not for boys ($b = 8.06$, $t(112) = .62$, $p = .53$). Specifically, girls endorsing low levels of disengagement coping had significantly fewer anxious/depressed symptoms than girls endorsing high levels of disengagement coping, whereas disengagement coping was unrelated to symptoms in boys (Figure 3).

A final full model was run including age, gender, and coping predicting symptoms. Notably, there was a significant three-way interaction between gender, age, and secondary control coping ($\beta = -8.67$, $p = .01$). Specifically, for boys, the association between secondary control coping and anxious/depressed symptoms was significant at *younger* ages ($b = -41.39$, $t(109) = -3.69$, $p < .001$), such that boys reporting more secondary control coping had fewer symptoms. In contrast, for girls, the relation among secondary control coping and symptoms was significant at *older* ages ($b = -63.50$, $t(109) = -4.68$, $p < .001$), where girls reporting more secondary control coping had fewer symptoms.

DISCUSSION

The current study examined the associations of youths' age, gender, and stress with coping and symptoms of anxiety and depression. Using a well-established control-based model of coping, findings showed significant age and gender differences in coping and internalizing symptoms. Findings build on prior research by further investigating associations with peer stress and demonstrating gender differences in the associations between specific types of coping strategies and internalizing symptoms across late childhood and early adolescence.

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Preliminary analyses replicated well-established age and gender differences in symptoms of anxiety and depression (e.g., Copeland et al., 2011; Hankin et al., 1998; Jalnapurkar et al., 2018; Salk et al., 2017). Further, findings demonstrated that anxious/depressed symptoms were associated with older age, specifically at high levels of peer stress. This replicates seminal work on contextual models of age differences in coping and research linking peer stress in adolescence with a heightened risk for internalizing disorders (Folkman et al., 1987; Folkman & Lazarus, 1980; Grant et al., 2003; McCrae, 1982; Zandstra et al., 2015; Zimmer-Gembeck, 2017).

Consistent with Hypothesis 1b and prior work, there was a positive association between secondary control coping and age. This may reflect improved executive functioning skills and a developmental shift in youths' capacity to use more cognitively complex coping strategies (e.g., cognitive reappraisal) (Anderson, 2002; Compas et al., 2017). In addition, consistent with prior research suggesting that the use of specific primary control coping strategies (e.g., support-seeking, problem-solving) increases with age (Zimmer-Gembeck & Skinner, 2011, 2017), youths' reports of primary control coping in the present sample were also associated with older age. However, the hypothesized association between age and disengagement coping was not supported.

In support of Hypothesis 2a/b, higher stress was associated with decreased use of primary and secondary control coping. Although primary and secondary control coping have shown associations with better psychological adjustment, youth encountering greater levels of stress tend to use fewer of these coping strategies (e.g., Wadsworth & Compas, 2002). The experience of chronic stress disrupts functioning in brain regions that are responsible for coping and self-regulation, which may, in turn, tax coping resources (Compas, 2006). Thus, it is possible that youth who experience heightened peer stress have a diminished capacity to utilize more traditionally effective forms of coping.

O*lder* youth who encountered lower levels of stress reported using fewer disengagement coping strategies. While it was expected that associations would emerge for heightened levels of stress (Hypothesis 3), it is possible that youth who reported less stress have a greater capacity to use more cognitively complex and active strategies (e.g., primary and secondary control coping). In light of findings that primary and secondary control coping were associated with older age, it is possible that older youth in the present sample reported decreased disengagement coping as a function of their propensity to use other coping strategies. Alternatively, it is also possible that youth who utilize less disengagement coping strategies may be encountering less stress.

While support was not found for primary control coping (Hypothesis 4), the three-way interaction between age, gender, and coping demonstrated a negative association between secondary control coping and anxious/depressed symptoms. This relationship was significant for boys at younger ages and girls at older ages. There are several possible interpretations of this finding. As analyses were unable to test secondary control coping at the strategy level, it is unclear if boys and girls in the present sample utilized different subtypes of secondary control coping strategies. For example, prior research suggests that boys may use more distraction, a form of secondary control coping, compared to girls (e.g.,

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Connor-Smith et al., 2000; Hampel & Petermann, 2005). Distraction is a less cognitively complex coping strategy and may be more readily employed by younger boys, potentially providing a buffer against the negative influence of peer stress. Older girls have been found to use more acceptance and cognitive reappraisal, which coincides with an increase in executive functioning skills (Anderson, 2002; Compas et al., 2017; Nolen-Hoeksema & Aldao, 2011). Therefore, it is possible that secondary control coping exhibits a buffering effect at different times in development for boys and girls. However, these interpretations should be assessed in longitudinal studies.

In support of the fifth hypothesis, older youth who relied more heavily on disengagement coping exhibited more internalizing symptoms, though this was significant only for girls. These findings are consistent with research identifying ineffective coping as a potential mechanism underlying gender differences in internalizing symptoms (e.g., Cyranowski et al., 2000). Further, the present study builds upon prior research by suggesting that internalizing problems associated with disengagement coping use may be exacerbated by additional risk factors (e.g., older age). Disengagement coping strategies may be most harmful for girls as they enter a period of heightened risk for internalizing problems during adolescence (Salk et al., 2017). Given findings that the use of secondary control coping corresponded to fewer symptoms for older girls, interventions that increase the use of these strategies (e.g., cognitive reappraisal, acceptance) while decreasing disengagement coping use (e.g., wishful thinking, avoidance, denial) may be particularly beneficial for girls.

Limitations and Future Directions

The present study had limitations that can be addressed in future research. First, the current was cross-sectional, and therefore, inferences concerning directionality and causal inferences cannot be made. For example, it could be that the presence of internalizing symptoms interferes with the use of particular coping strategies or, conversely, that use of these coping strategies leads to reductions in later symptoms. Longitudinal studies will be important to assess developmental changes in coping, associations with psychopathology over time, and the temporal precedence of youths' coping, stress, and symptoms. Second, the present study used a single-informant and self-report design. It is possible that the magnitude of associations may reflect shared method variance to some degree (e.g., LaGrange & Cole, 2008). However, in a meta-analysis by Compas and colleagues (2017), informant (e.g., single vs. multiple-informant) was tested as a moderator of the relations among coping factors and strategies with internalizing and externalizing problems in cross-sectional studies, yielding few significant effects. Third, the modest sample size of the present study was adequately powered to detect small-to-medium effects. Future research should examine associations among key study variables in larger samples to replicate and extend research on developmental and gender differences in coping and psychopathology. Last, the present study used a control-based model that delineates coping at the factor level (e.g., secondary control coping) (Compas et al., 2017; Connor-Smith et al., 2000; Rudolph et al., 1995; Weisz et al., 1994). Future research should examine potential differences at the strategy level (e.g., distraction vs. cognitive reappraisal). This would enable interventionists to discern if teaching specific coping strategies to boys and girls at different points of development would be the most effective way to reduce the risk of internalizing disorders.

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Conclusion

Taken together, the current research suggests that there are significant gender- and age-related differences in coping and symptoms of internalizing psychopathology. Findings showed a positive association between the use of more cognitively complex and active coping strategies with age and extend prior research by showing gender differences in the relation among secondary control coping use and anxious/depressed symptoms at different ages. Further, findings show that peer stress exposure may relate to disengagement coping use across development. The present results provide support for interventions that target increased use of primary and secondary control coping strategies to prevent internalizing problems in youth.

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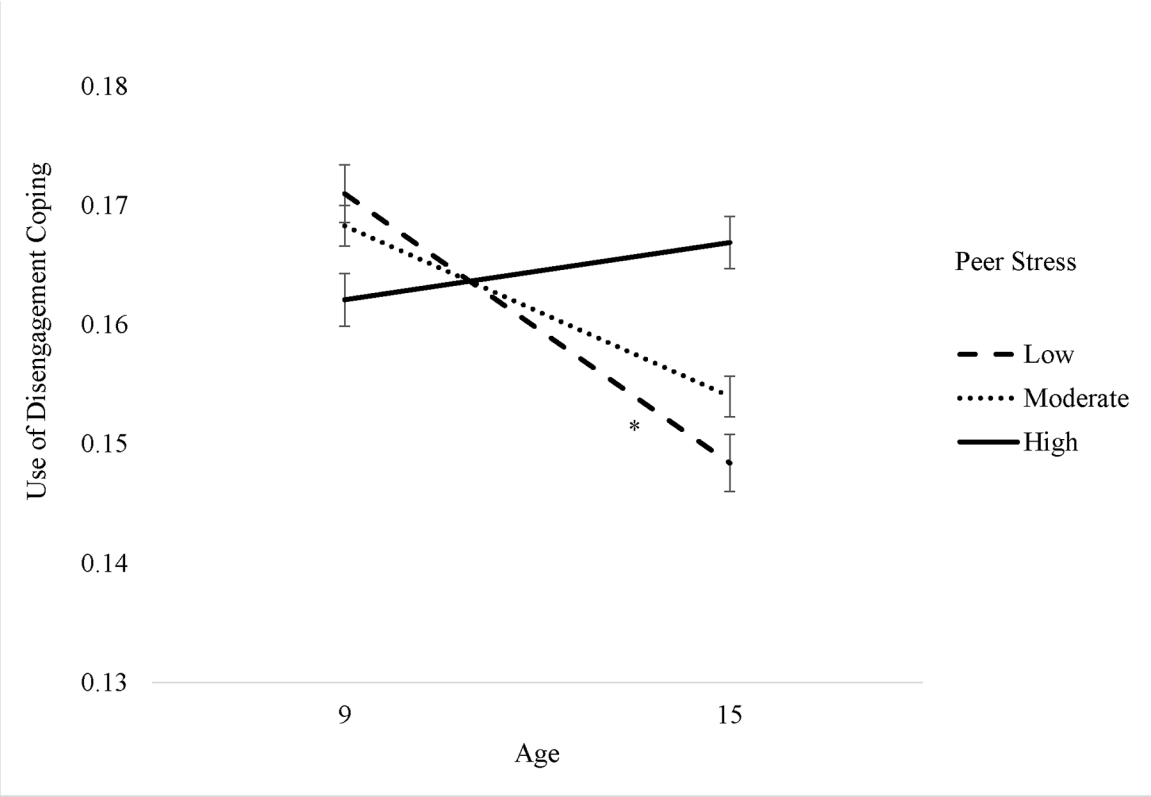

Figure 1.
Interaction Between Age and Stress Predicting Youths' Use of Disengagement Coping

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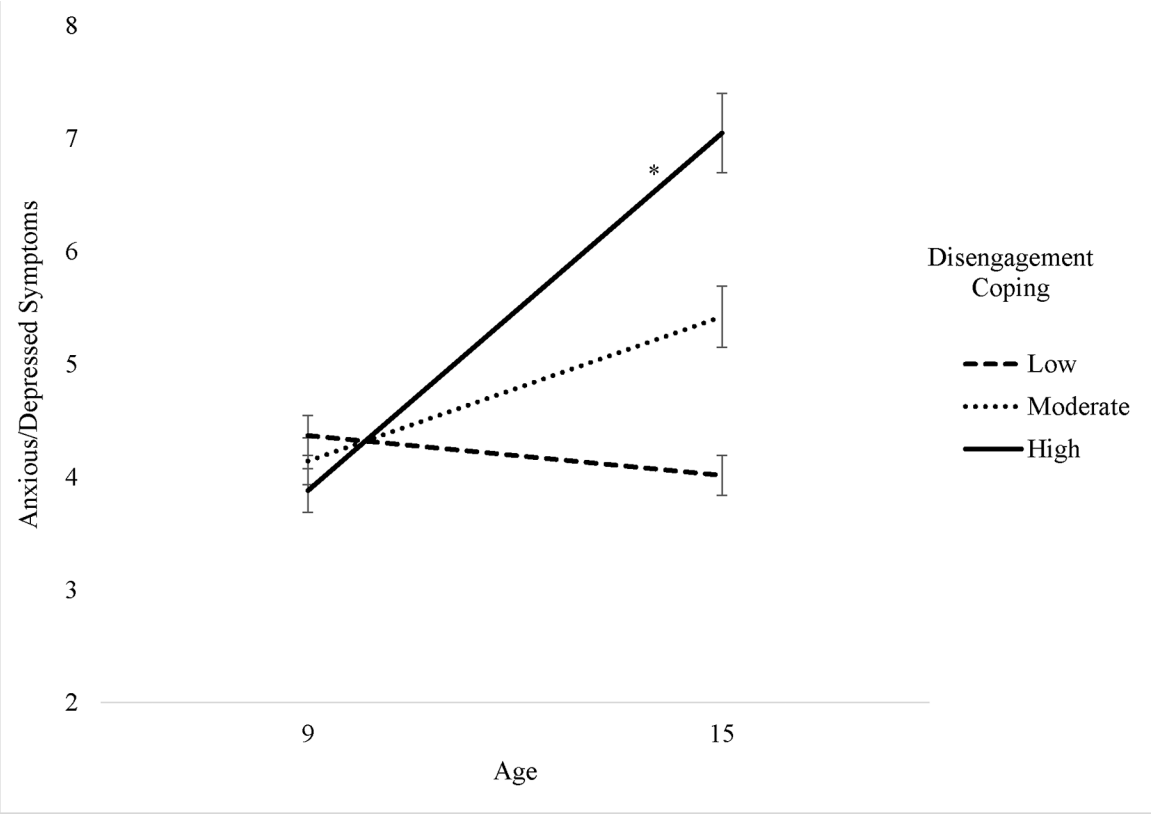

Figure 2.

Interaction Between Age and Disengagement Coping Predicting Youths' Anxious/Depressed Symptoms

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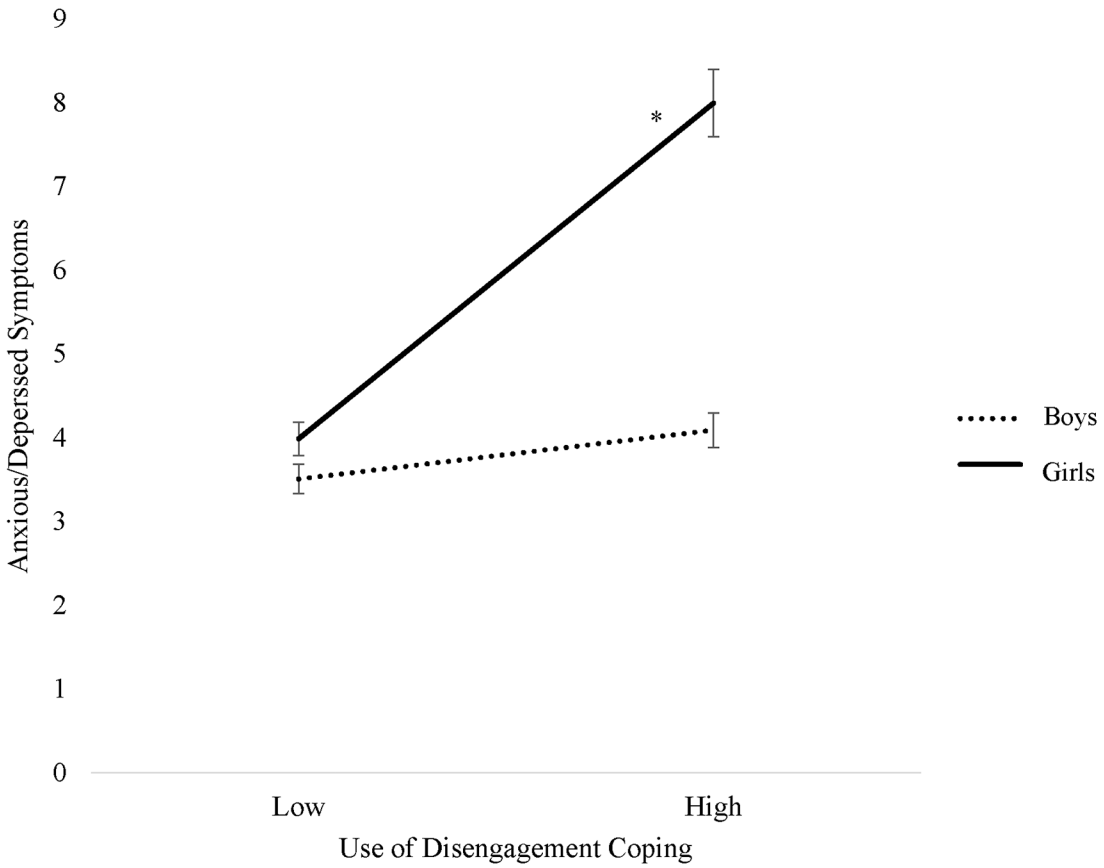

Figure 3.
Interaction Between Gender and Disengagement Coping Predicting Youths' Anxious/Depressed Symptoms

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Table 1

Descriptive Statistics for Key Study Variables

Boys (*n* = 63)	***M***	***SD***	**Min**	**Max**
Age	12.44	1.82	9	15
Peer Stress	15.93	5.80	10	30
Anxious/Depressed Symptoms	3.83	3.60	0	15
Primary Control Coping	.18	.05	.08	.30
Secondary Control Coping	.26	.06	.11	.39
Disengagement Coping	.16	.04	.10	.23
Girls (*n* = 54)	***M***	***SD***	**Min**	**Max**
Age	12.06	2.00	9	15
Peer Stress	16.20	5.93	10	34
Anxious/Depressed Symptoms	5.91	4.17	0	18
Primary Control Coping	.19	.04	.09	.31
Secondary Control Coping	.25	.05	.16	.36
Disengagement Coping	.16	.03	.09	.24

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Table 2

Bivariate Correlation Matrix Among Key Study Variables

	1	2	3	4	5
1. Age	--				
2. Stress	.04	--			
3. Anxious/Depressed Symptoms	.09	.55***	--		
4. Primary Control Coping	.16	−.20*	−.20*	--	
5. Secondary Control Coping	.21*	−.43***	−.45***	−.20*	--
6. Disengagement Coping	−.12	.09	.23*	−.58***	−.20*

Note.

* $p < .05$

** $p < .01$

*** $p < .001$

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Table 3

Linear Regression Analyses Predicting Youths' Coping from Age, Gender, and Stress.

Model and Predictor	Primary Control Coping		Secondary Control Coping		Disengagement Coping	
	F	**Adjusted R^2**	**F**	**Adjusted R^2**	**F**	**Adjusted R^2**
Model 1	2.74*	.06*	9.65***	.23***	2.45	.05
		t		*t*		*t*
Gender	.15	1.60	−.03	−.31	−.01	−.11
Age	.18	1.95	.22	2.73**	−.13	−1.41
Stress	−.21	−2.36*	−.45	−5.51***	.12	1.34
Age × Stress	−.10	−1.10	−.14	−1.65	.25	2.64**
	F	**Adjusted R^2**	**F**	**Adjusted R^2**	**F**	**Adjusted R^2**
Model 2	1.83	.02	3.71*	.07*	.65	−.01
		t		*t*		*t*
Gender	.12	1.33	−.07	−.72	.05	.48
Age	.18	1.92	.23	2.58*	−.12	−1.24
Age × Gender	−.08	−.91	−.20	−2.21*	.04	.40
	F	**Adjusted R^2**	**F**	**Adjusted R^2**	**F**	**Adjusted R^2**
Model 3	1.74	.04	5.90***	.23***	1.52	.03
		t		*t*		*T*
Gender	.15	1.55	−.03	−.30	−.01	−.09
Age	.20	2.16*	.22	2.62*	−.12	−1.31
Stress	−.22	−2.24*	−.43	−5.00***	.13	1.37
Age × Gender	−.05	−.49	−.12	−1.44	.04	.42
Age × Stress	−.09	−.94	−.17	−1.90	.27	2.75**
Gender × Stress	−.07	−.70	.08	.91	−.07	−.74
Age × Stress × Gender	.07	.72	.00	.03	−.06	−.65

Note.

*$p < .05$

**$p < .01$

*** $p < .001$

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Table 4

Linear Regression Analyses Predicting Youths' Anxious/depressed Symptoms from Age, Coping, and Gender.

Model 1			**Model 4**		
	F	**Adjusted R^2**		**F**	**Adjusted R^2**
	4.78**	.12**		5.15***	.13***
		T			*T*
Gender	.31	3.42***	Age	.16	1.80
Age	.16	1.78	Gender	.31	3.52***
PCC	−.26	−2.64**	PCC	.02	−.08
Age × PCC	−.01	−.06	Gender × PCC	−.30	−1.13
	F	**Adjusted R^2**		**F**	**Adjusted R^2**
Model 2	12.12***	.28***	**Model 5**	12.64***	.29***
		T			*T*
Gender	.25	3.11**	Age	.20	2.46*
Age	.23	2.73**	Gender	.24	3.01**
SCC	−.48	−5.94***	SCC	−.17	−.71
Age × SCC	.05	.61	Gender × SCC	−.32	−1.35
	F	**Adjusted R^2**		**F**	**Adjusted R^2**
Model 3	6.47***	.16***	**Model 6**	5.89***	.14***
		T			*T*
Gender	.26	3.00**	Age	.14	1.64
Age	.13	1.44	Gender	.26	3.02**
DC	.20	2.34*	DC	−.34	−1.29
Age × DC	.23	2.71**	Gender × DC	.60	2.31*

Note. PCC = Primary Control Coping; SCC = Secondary Control Coping; DC = Disengagement Coping.

* $p < .05$

** $p < .01$

*** $p < .001$